



# Continuum Mechanics

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# Chapter 01

## Tensor Algebra

### Part 4 of 5



# Eigenvalues, Eigenvectors, and Principal Invariants

# Eigenvalues and Eigenvectors



## Definition

By an *eigenpair* for a second-order tensor  $S \in \mathcal{V}^2$ , we mean a scalar  $\lambda \in \mathbb{R}$  and a unit vector  $\mathbf{e} \in \mathcal{V}$  satisfying

$$S\mathbf{e} = \lambda\mathbf{e}.$$

Any such  $\lambda$  is called an *eigenvalue* of  $S$ , and any such  $\mathbf{e}$  is called an *eigenvector* of  $S$ .



## Proposition

Let  $S \in \mathcal{V}^2$  be given. If  $\lambda \in \mathbb{R}$  is an eigenvalue of  $S$ , then

$$p(\lambda) = 0,$$

where

$$p(\alpha) = \det(S - \alpha I), \quad \alpha \in \mathbb{R}. \quad (1.15)$$

The polynomial  $p$  is called the characteristic polynomial of  $S$ .



## Proposition

*Suppose that  $S \in \mathcal{V}^2$  is given and  $\lambda \in \mathbb{R}$  is an eigenvalue of  $S$ . A solution of*

$$(S - \lambda I)\mathbf{e} = \mathbf{0} \quad (1.16)$$

*yields non-unique eigenvectors of  $S$  associated with  $\lambda$ .*

Indeed, every nonzero solution of (1.16) can be normalized, and multiplication of an eigenvector by  $-1$  gives another unit eigenvector for the same eigenvalue.



## Remark

*Here we have decided to ignore complex eigenvalues, not because they are unimportant, but because they are not as useful in continuum mechanics.*

*The solutions of*

$$p(\lambda) = 0$$

*could occur in complex-conjugate pairs.*



## Result 1.8

- ① The eigenvalues of an SPD tensor are strictly positive.
- ② Any two eigenvectors corresponding to distinct eigenvalues of a symmetric tensor are orthogonal.



## Proof.

For part (1), let  $S \in \mathcal{V}^2$  be SPD and let  $(\lambda, \mathbf{e})$  be an eigenpair. Then

$$S\mathbf{e} = \lambda\mathbf{e}.$$

Because  $\mathbf{e}$  is a unit vector,

$$\begin{aligned} 0 &< \mathbf{e} \cdot (S\mathbf{e}) \\ &= \mathbf{e} \cdot (\lambda\mathbf{e}) \\ &= \lambda(\mathbf{e} \cdot \mathbf{e}) \\ &= \lambda. \end{aligned}$$

Thus  $\lambda > 0$ .



## Proof (Cont.)

For part (2), suppose that  $S$  is symmetric and that  $(\lambda, \mathbf{w})$  and  $(\mu, \mathbf{u})$  are eigenpairs. Assume  $\lambda \neq \mu$ . Then

$$\begin{aligned}\lambda(\mathbf{u} \cdot \mathbf{w}) &= \mathbf{u} \cdot (\lambda \mathbf{w}) \\ &= \mathbf{u} \cdot (S\mathbf{w}) \\ &= \mathbf{w} \cdot (S\mathbf{u}) \quad (\text{symmetry}) \\ &= \mathbf{w} \cdot (\mu \mathbf{u}) \\ &= \mu(\mathbf{w} \cdot \mathbf{u}).\end{aligned}$$

Hence

$$(\lambda - \mu)(\mathbf{w} \cdot \mathbf{u}) = 0.$$

Since  $\lambda \neq \mu$ , cancellation gives

$$\mathbf{w} \cdot \mathbf{u} = 0.$$





## Result 1.9 (Spectral Decomposition Theorem)

Let  $S \in \mathcal{V}^2$  be symmetric. There is a right-handed orthonormal frame

$$\mathcal{F} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$$

consisting of eigenvectors of  $S$ . The corresponding eigenvalues  $\lambda_i$ ,  $i = 1, 2, 3$ , are the same, up to ordering, for any such frame and form the full set of eigenvalues of  $S$ .

The dyadic representation of  $S$  in any such frame is

$$S = \sum_{i=1}^3 \lambda_i \mathbf{e}_i \otimes \mathbf{e}_i,$$

and its matrix representation is

$$[S] = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}.$$



## Result 1.9 (Cont.)

The three eigenvalues are unique up to ordering, even when an eigenspace has dimension greater than one. Within a repeated-eigenvalue eigenspace, the orthonormal eigenvectors need not be unique.

## Proof.

A standard result from linear algebra. □



# Principal Invariants

## Definition

Let  $S \in \mathcal{V}^2$  be arbitrary. The *principal invariants* of  $S$  are the three scalar quantities

$$I_1(S) = \text{tr}(S),$$

$$I_2(S) = \frac{1}{2} [(\text{tr}(S))^2 - \text{tr}(S^2)],$$

$$I_3(S) = \det(S).$$



## Proposition

Let  $S \in \mathcal{V}^2$  be a given tensor and  $\alpha \in \mathbb{R}$  a scalar. Then

$$\det(S - \alpha I) = -\alpha^3 + l_1(S)\alpha^2 - l_2(S)\alpha + l_3(S) = p(\alpha).$$

## Proof.

Let  $\mathcal{F} = \{\mathbf{e}_i\}$  be a frame. Compute the characteristic polynomial  $p(\alpha) = \det([S] - \alpha[I])$  and compare coefficients with the definitions of  $l_1(S)$ ,  $l_2(S)$ , and  $l_3(S)$ . □



## Proposition

Let  $S \in \mathcal{V}^2$  be symmetric, with eigenvalues  $\lambda_1, \lambda_2, \lambda_3$ . Then

$$I_1(S) = \lambda_1 + \lambda_2 + \lambda_3,$$

$$I_2(S) = \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1,$$

$$I_3(S) = \lambda_1\lambda_2\lambda_3.$$

Proof.

Exercise. □



## Remark

*The quantities  $I_1$ ,  $I_2$ , and  $I_3$  are called invariants because they are built from the invariant quantities  $\text{tr}(S)$  and  $\det(S)$ . See Results 1.5 and 1.6.*



## Result 1.10 (Cayley–Hamilton Theorem)

Let  $S \in \mathcal{V}^2$  be arbitrary. Then

$$S^3 - I_1(S)S^2 + I_2(S)S - I_3(S)I = O.$$

Proof.

Exercise. □



# Special Decompositions



## Result 1.11 (The SPD Square Root)

Let  $C \in \mathcal{V}^2$  be SPD. There is a unique SPD tensor  $U$  such that

$$U^2 = C.$$

In particular,

$$U = \sum_{i=1}^3 \sqrt{\lambda_i} \mathbf{e}_i \otimes \mathbf{e}_i,$$

where  $\mathcal{F} = \{\mathbf{e}_i\}$  is any frame of eigenvectors of  $C$  and  $\lambda_i > 0$  are the corresponding eigenvalues of  $C$ . We will write, in this case,  $U = \sqrt{C}$ .

Proof.

Exercise. □



## Result 1.12 (Polar Decomposition)

Let  $F \in \mathcal{V}^2$  satisfy

$$\det(F) > 0.$$

Then there exist right and left polar decompositions of  $F$  of the forms, respectively,

$$F = RU \quad \text{and} \quad F = VR,$$

where

$$U = \sqrt{F^T F} \quad \text{and} \quad V = \sqrt{F F^T}$$

are both SPD, and  $R$  is a rotation.



## Proof.

Since  $\det(F) > 0$ , the tensor  $F$  is invertible. Thus

$$F\mathbf{w} \neq \mathbf{0} \quad \text{for every } \mathbf{w} \in \mathcal{V}_* := \mathcal{V} \setminus \{\mathbf{0}\}.$$

Similarly,  $F^T$  is invertible and

$$F^T \mathbf{w} \neq \mathbf{0} \quad \text{for every } \mathbf{w} \in \mathcal{V}_*.$$

Therefore,

$$\mathbf{w} \cdot (F^T F \mathbf{w}) = (F \mathbf{w}) \cdot (F \mathbf{w}) > 0,$$

$$\mathbf{w} \cdot (F F^T \mathbf{w}) = (F^T \mathbf{w}) \cdot (F^T \mathbf{w}) > 0,$$

for every  $\mathbf{w} \in \mathcal{V}_*$ .



## Proof (Cont.)

Thus  $F^T F$  and  $FF^T$  are SPD. Using Result 1.11, define

$$U := \sqrt{F^T F}, \quad V := \sqrt{FF^T}.$$

Next, define

$$R := FU^{-1}.$$

Then  $RU = F$ . Observe that

$$\begin{aligned} 0 < \det(F) &= \det(RU) \\ &= \det(R) \det(U). \end{aligned}$$

Also, since  $U$  is SPD,

$$\det(U) > 0.$$

Therefore  $\det(R) > 0$ .



## Proof (Cont.)

Furthermore,

$$\begin{aligned} R^T R &= U^{-T} F^T F U^{-1} \\ &= U^{-1} U^2 U^{-1} \\ &= I, \end{aligned}$$

because  $U^T = U$ . Hence  $R$  is orthogonal, so

$$|\det(R)| = 1.$$

Together with  $\det(R) > 0$ , this gives  $\det(R) = 1$ . Thus  $R$  is a rotation, and the right polar decomposition

$$F = RU$$

is proved.



## Proof (Cont.)

To obtain the left polar decomposition, define

$$Q := V^{-1}F.$$

The same calculation used for  $R$  shows that  $Q$  is a rotation. Hence

$$RU = F = VQ.$$

Let

$$S := Q^T VQ.$$

Because  $V$  is SPD and  $Q$  is orthogonal,  $S$  is SPD. Indeed,  $S^T = S$ , and for  $\mathbf{w} \neq \mathbf{0}$ ,

$$\mathbf{w} \cdot (S\mathbf{w}) = (Q\mathbf{w}) \cdot (VQ\mathbf{w}) > 0.$$



## Proof (Cont.)

Using  $RU = VQ$ ,

$$\begin{aligned} RU &= VQ \\ &= QQ^T VQ \\ &= QS, \end{aligned}$$

which implies

$$R = QSU^{-1}.$$

Since  $R$  and  $Q$  are rotations,

$$R^T R = I, \quad Q^T Q = I.$$

Also,  $S^T = S$  and  $U^T = U$  because  $S$  and  $U$  are SPD.



## Proof (Cont.)

Using  $R = QSU^{-1}$  and the identities from the previous slide,

$$\begin{aligned} I &= R^T R \\ &= (QSU^{-1})^T (QSU^{-1}) \\ &= U^{-T} S^T Q^T Q S U^{-1} \\ &= U^{-1} S^2 U^{-1}. \end{aligned}$$

Multiplying this equality on the left and on the right by  $U$  gives

$$U^2 = S^2.$$

Since  $U^2 = F^T F$  by definition,

$$S^2 = U^2 = F^T F.$$

Thus  $S$  and  $U$  are both SPD square roots of  $F^T F$ .



## Proof (Cont.)

By uniqueness of the SPD square root,  $U = S$ . Hence

$$R = QSU^{-1} = Q.$$

Therefore,

$$RU = F = VR.$$

This gives both polar decompositions with the same rotation tensor  $R$ . □



# Scalar Product for Second-Order Tensors



# Tensor Inner Product

## Definition

Let  $S, D \in \mathcal{V}^2$ . The *tensor inner product* is defined by

$$S : D := \text{tr}(S^T D). \quad (1.17)$$



## Proposition

Let  $\mathcal{F} = \{\mathbf{e}_i\}$  be a frame. Then the dyadic basis for  $\mathcal{V}^2$ ,

$$\{\mathbf{e}_i \otimes \mathbf{e}_j \mid 1 \leq i, j \leq 3\},$$

is orthonormal with respect to the tensor inner product; that is,

$$(\mathbf{e}_i \otimes \mathbf{e}_j) : (\mathbf{e}_k \otimes \mathbf{e}_\ell) = \delta_{ik} \delta_{j\ell}.$$

Proof.

Exercise. □



## Result 1.13

Let  $\mathcal{F} = \{\mathbf{e}_i\}$  be a frame. Suppose

$$S = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j, \quad D = D_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Then

$$S : D = S_{ij} D_{ij}.$$

If  $S$  is symmetric, then

$$S : D = S : \text{sym}(D) = \text{sym}(S) : \text{sym}(D).$$



## Proof.

Using the dyadic expansions and orthonormality of the dyadic basis,

$$\begin{aligned} S : D &= (S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j) : (D_{k\ell} \mathbf{e}_k \otimes \mathbf{e}_\ell) \\ &= S_{ij} D_{k\ell} (\mathbf{e}_i \otimes \mathbf{e}_j) : (\mathbf{e}_k \otimes \mathbf{e}_\ell) \\ &= S_{ij} D_{k\ell} \delta_{ik} \delta_{j\ell} \\ &= S_{ij} D_{ij}. \end{aligned}$$

Here all repeated indices are dummy indices; the contractions with  $\delta_{ik}$  and  $\delta_{j\ell}$  leave no free index.



## Proof (Cont.)

If  $S$  is symmetric, then  $S_{ij} = S_{ji}$ . Therefore,

$$\begin{aligned} S : D &= S_{ij} D_{ij} \\ &= \frac{1}{2} (S_{ij} D_{ij} + S_{ji} D_{ji}) \\ &= \frac{1}{2} S_{ij} (D_{ij} + D_{ji}) \\ &= S_{ij} [\text{sym}(D)]_{ij} \\ &= S : \text{sym}(D). \end{aligned}$$

Since  $\text{sym}(S) = S$ , it follows that

$$S : D = \text{sym}(S) : \text{sym}(D).$$





# Fourth-Order Tensors



## Fourth-Order Tensors

### Definition

A *fourth-order tensor* is a mapping

$$\mathbf{C} : \mathcal{V}^2 \longrightarrow \mathcal{V}^2$$

that is linear in the sense that

$$\begin{aligned}\mathbf{C}(S + T) &= \mathbf{C}(S) + \mathbf{C}(T), & S, T \in \mathcal{V}^2, \\ \mathbf{C}(\alpha T) &= \alpha \mathbf{C}(T), & \alpha \in \mathbb{R}, \quad T \in \mathcal{V}^2.\end{aligned}$$

The set of all fourth-order tensors is denoted  $\mathcal{V}^4$ .



## Zero, Identity, and Equality

The *zero fourth-order tensor*  $\mathbf{O} \in \mathcal{V}^4$  is characterized by

$$\mathbf{O}S = \mathbf{O}, \quad S \in \mathcal{V}^2.$$

The *identity fourth-order tensor*  $\mathbf{I} \in \mathcal{V}^4$  is characterized by

$$\mathbf{I}S = S, \quad S \in \mathcal{V}^2.$$

Two fourth-order tensors  $\mathbf{C}, \mathbf{D} \in \mathcal{V}^4$  are called *equal* if and only if

$$\mathbf{C}T = \mathbf{D}T, \quad T \in \mathcal{V}^2.$$



## Example

Fix  $A \in \mathcal{V}^2$ . Define

$$\mathbf{C}(T) = AT, \quad T \in \mathcal{V}^2.$$

Then  $\mathbf{C}$  is a fourth-order tensor, because

$$\mathbf{C}(S + T) = A(S + T) = AS + AT = \mathbf{C}(S) + \mathbf{C}(T),$$

$$\mathbf{C}(\alpha T) = A(\alpha T) = \alpha AT = \alpha \mathbf{C}(T).$$



## Proposition

Let  $\mathbf{C}, \mathbf{D} \in \mathcal{V}^4$  and  $\alpha, \beta \in \mathbb{R}$ . Define

$$(\alpha \mathbf{C} + \beta \mathbf{D})T := \alpha \mathbf{C}(T) + \beta \mathbf{D}(T), \quad T \in \mathcal{V}^2.$$

Define the product  $\mathbf{CD} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$  by

$$(\mathbf{CD})(T) := \mathbf{C}(\mathbf{D}(T)), \quad T \in \mathcal{V}^2.$$

Then  $\mathcal{V}^4$  is a linear vector space. Furthermore,  $\mathbf{CD} \in \mathcal{V}^4$ .

## Proof.

An easy exercise using the linearity of  $\mathbf{C}$  and  $\mathbf{D}$ . □



# Representation in a Coordinate Frame



# Components of a Fourth-Order Tensor

## Definition

Let  $\mathbf{C} \in \mathcal{V}^4$  and suppose  $\mathcal{F} = \{\mathbf{e}_i\}$  is a right-handed orthonormal frame. The 81 components of  $\mathbf{C}$ ,

$$C_{ijkl} \in \mathbb{R}, \quad 1 \leq i, j, k, \ell \leq 3,$$

are defined by

$$C_{ijkl} = \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell)\mathbf{e}_j).$$



## Proposition

Suppose that  $\mathbf{C} \in \mathcal{V}^4$  and  $U, T \in \mathcal{V}^2$  satisfy

$$U = \mathbf{C}(T).$$

Let  $\mathcal{F} = \{\mathbf{e}_i\}$  be a frame and write

$$U = U_{ij} \mathbf{e}_i \otimes \mathbf{e}_j, \quad T = T_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Then

$$U_{ij} = C_{ijkl} T_{kl}. \quad (1.18)$$

Here  $i, j$  are free indices, while  $k, \ell$  are dummy indices.



## Proof.

We have

$$\begin{aligned} U_{ij} &= \mathbf{e}_i \cdot (U \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (\mathbf{C}(T) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (\mathbf{C}(T_{k\ell} \mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) T_{k\ell} \\ &= C_{ijk\ell} T_{k\ell}. \end{aligned}$$

The free indices  $i, j$  agree in every line, and  $k, \ell$  are dummy indices. □



## Example

Let  $A \in \mathcal{V}^2$  be given and define  $\mathbf{C} \in \mathcal{V}^4$  by

$$\mathbf{C}(T) = AT.$$

Then

$$\begin{aligned} C_{ijkl} &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot ((A\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (A\mathbf{e}_k (\mathbf{e}_\ell \cdot \mathbf{e}_j)) \\ &= \delta_{\ell j} \mathbf{e}_i \cdot (A\mathbf{e}_k) \\ &= A_{ik} \delta_{\ell j}. \end{aligned}$$



## Proposition

Let  $\mathbf{C} \in \mathcal{V}^4$ . Then its components may equivalently be written

$$C_{ijkl} = (\mathbf{e}_i \otimes \mathbf{e}_j) : \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l).$$



## Proof.

We first show that, for any  $A \in \mathcal{V}^2$ ,

$$A_{ij} = \mathbf{e}_i \cdot (A\mathbf{e}_j) = (\mathbf{e}_i \otimes \mathbf{e}_j) : A.$$

Indeed,

$$\begin{aligned}\mathbf{e}_i \otimes \mathbf{e}_j &= \delta_{pi} \delta_{qj} \mathbf{e}_p \otimes \mathbf{e}_q, & (\text{sum on } p, q), \\ (\mathbf{e}_i \otimes \mathbf{e}_j) : A &= \delta_{pi} \delta_{qj} (\mathbf{e}_p \otimes \mathbf{e}_q) : A \\ &= \delta_{pi} \delta_{qj} A_{pq} \\ &= A_{ij},\end{aligned}$$

where Result 1.13 is applied to each dyad  $\mathbf{e}_p \otimes \mathbf{e}_q$ . Here  $i, j$  are free, while  $p, q$  are dummy.



## Proof (Cont.)

For fixed  $k, \ell$ , define

$$A^{(k\ell)} := \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \in \mathcal{V}^2.$$

Applying the identity from the previous slide to  $A^{(k\ell)}$  gives

$$\begin{aligned} C_{ijk\ell} &= \mathbf{e}_i \cdot (\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell) \mathbf{e}_j) \\ &= \mathbf{e}_i \cdot (A^{(k\ell)} \mathbf{e}_j) \\ &= (\mathbf{e}_i \otimes \mathbf{e}_j) : A^{(k\ell)} \\ &= (\mathbf{e}_i \otimes \mathbf{e}_j) : \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_\ell), \end{aligned}$$

which proves the result. □